



THE WHALE IS BACK AND IT IS HUNGRY! MIRACULOUS SEASONAL PATTERNS IN THE PASSIVE ERA

- 1 – Target-date funds must rebalance into stocks at the end of the month / quarter after large corrections
- 2 – Stocks have rallied by an absurd 80.6% p.a. in the last week of the quarter during down quarters since 2016
- 3 – Assets which are not owned by TDFs are not impacted by this odd mean-reverting pattern
- 4 – The stock market's momentum anomaly has disappeared since the rise of TDFs
- 5 – TDF rebalancing may be responsible for the stonk bubble and the Fed's tardiness to tackle inflation

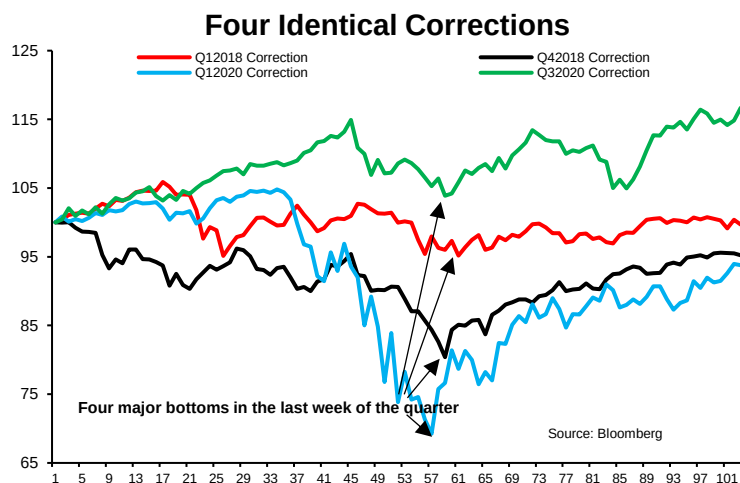
I would like to thank my friend Mike Green of Simplify ETFs for suggesting that I write this report and for his tireless advocacy on the distortions and risks created by the rise of passive investing and J. Parker, A. Scholar, and Y. Sun for their excellent paper, [Retail Financial Innovation and Stock Market Dynamics: the Case of Target Date Funds](#)

The whale is back! The S&P 500 index rebounded by a miraculous 3.8% on February 23rd and 24th despite the spike of Brent prices above \$100 a barrel, crippling sanctions against Russia, and the real possibility of nuclear war. This absurd end-of-month rally follows the pattern I described in [Swimming with the Target-Date Whale](#): “March 23, 2018. December 24, 2018. March 23, 2020. September 23, 2020. **The past four bear markets or corrections all ended with a V-shaped bottom on the exact same date**”.

This report will argue that the **rebalancing target-date funds creates abnormal seasonal patterns, nonsensical cross-asset correlations, and monetary policy blunders.**

The first part will explain the theory behind target-date funds rebalancing, as evidenced by the massive counter-cyclical flows in the largest equity mutual funds in recent corrections.

The second part will show that seasonal patterns confirm the target-date whale hypothesis: **stocks have rallied massively in the last week of the month / quarter when TDFs needed to rebalance since 2016.** This effect is not observed with assets which are not owned by TDFs, such as commodities and bitcoin.



The third part will discuss the macro dangers of the TDF whale. First, relentless buying by target-date funds stops bear markets, prevents valuations from normalizing, and **feed speculative frenzies.**

Second, TDF rebalancing magnifies the “wealth effect”, which will also **increase the pain of the upcoming rate hikes.**

Third, TDF rebalancing changes the correlation between stocks and bonds and may have **led the Fed to dismiss persistent inflationary signs,** resulting in the policy nightmare of hiking amidst an oil shock.



Theory

P. Volcker famously said that the most important financial innovation of the past two decades has been the ATM. He should have added index funds and target-date funds to that sentence, although their ultimate impact on society may not be as positive as that of the humble ATM. This report will show how **index-based target-date funds have transformed the structure of capital markets.**

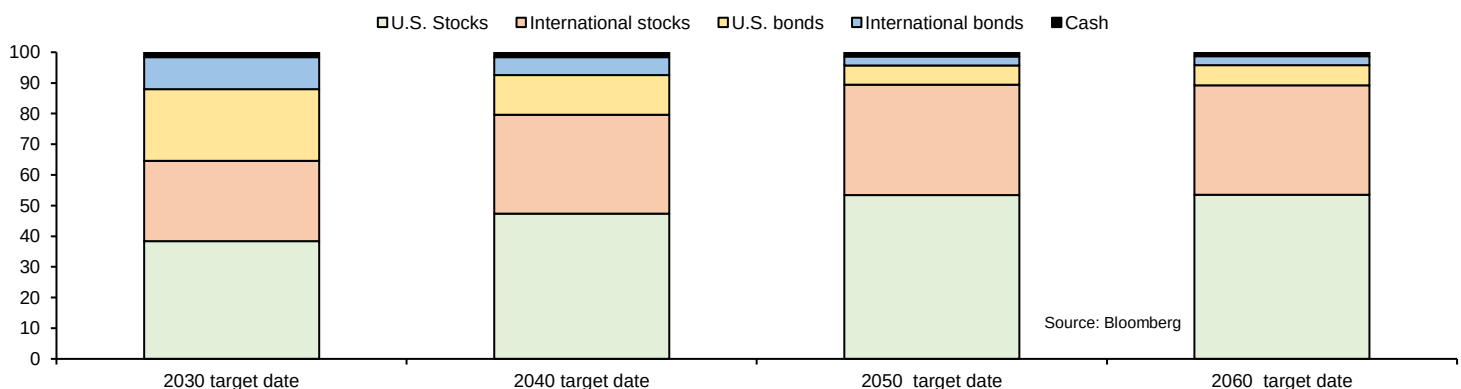
Target-date funds automate the retirement account work which humans used to do. The theoretical case for using index fund-based target-date funds is strong.

- As B. Sharpe argued, the active sector will underperform a market cap-weighted index by the amount spent on research and trading
- Retail investors are notoriously horrible market-timers
- Over time, “time in the market” and asset allocation are the most important drivers of performance
- Time smooths volatility and assets’ returns eventually converge towards their long-term averages. Young workers should stick to the plan and trust in the power of compounding
- If assets’ returns are mean-reverting (which they probably are or asset classes would eventually disappear), automatic rebalancing can generate a small excess return by buying low and selling high.

As a result, most U.S. workers are advised to invest 7 to 10% of their monthly paychecks in a tax-advantaged 401-k account and select a target-date fund which will minimize fees, automate contributions, and gently reduce risk as retirement approaches. The Pension Protection Act of 2006 cemented TDFs by making them **the most popular qualified default investment option in defined contribution retirement saving plans.** TDFs assets have risen from less than \$8 billion in 2000 **to close to \$3 trillion today** (of the roughly \$21 trillion in US mutual funds).

Index funds-based TDF allocation is remarkably simple. **The most popular TDF ETFs own just four index funds** for U.S. equities, international stocks, U.S. bonds, and international bonds. The weights assigned to each asset class is a function of the expected retirement date, which determine beneficiaries’ tolerance for risk.

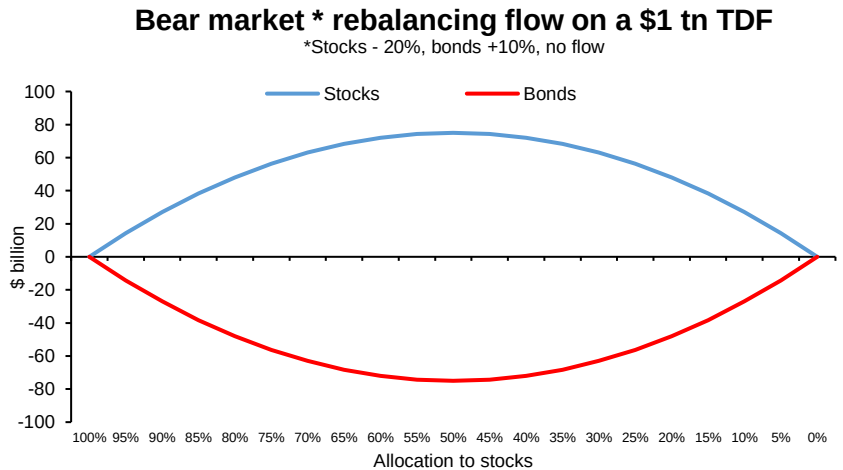
Asset Allocation of the Major Target-Date Funds





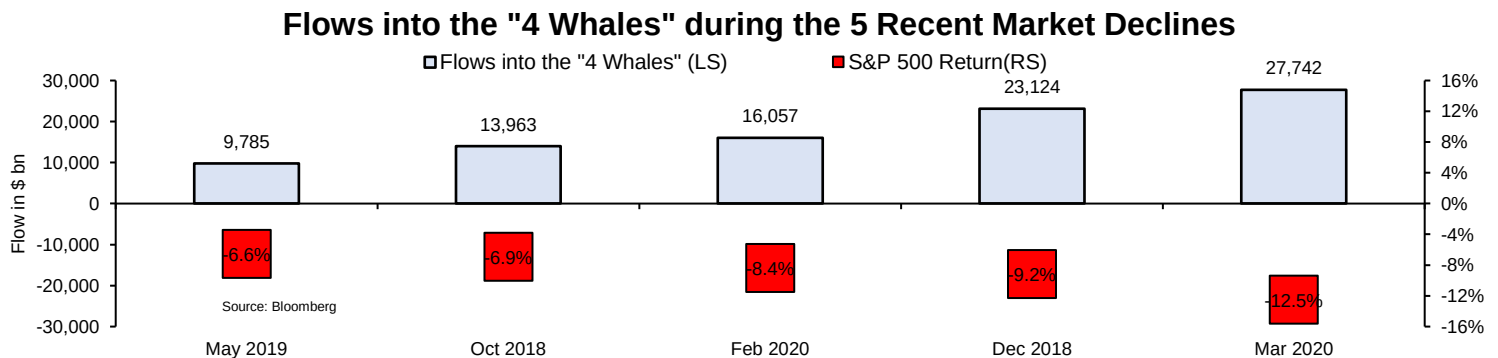
Target-date funds' trading activity is driven by only two considerations: flows and past performance. Inflows are usually invested promptly as TDFs try to maximize "time in the market" and minimize tracking error. Past performance-driven trades follow the iron rule of rebalancing: **what has gone down must be bought and what has gone up must be sold.**

The amount of rebalancing is a quadratic function of the proportion of the funds' allocation to each asset class. At an extreme, funds which are invested solely in stocks or solely in bonds never need to rebalance. **The need for rebalancing is greatest for funds which are perfectly balanced.** For example, the chart at right shows that a \$1 trillion with 50% stocks and 50% bonds needs to shift \$75 billion from the latter to the former to return to its target allocation after a run-of-the-mill bear market (stocks down 20%, bonds up by 10%)¹.



Target-date funds typically adjust their allocation with flows. Inflows are invested into the underperforming asset and redemptions are met by selling the outperforming asset. However, the rebalancing needs caused by extreme market moves exceed flows' ability to restore TDFs' policy allocation. In this case, TDFs need to actively rotate from one asset to the other. Since trading is costly and markets are fickle, TDFs usually wait until the end of the month or the quarter to get back to their policy weights.

The chart below shows the flows in the four largest equity funds (the "four whales" for short) in the five most recent big market corrections. **Each equity selloff was met with inflows: the larger the drawdown, the larger the inflow.** Any sell-side trader can confirm that this pattern violates the first rule of flows: clients' money follows performance. The \$27 billion (!) inflow in March 2020 is particularly stunning: locked-down investors were hit with margin calls and clients were selling in a panic. Only a purely quant-driven whale following a rebalancing algorithm could have bought so much stocks amidst this carnage.



¹ See J. Parker, A. Scholar, and Y. Sun, [Retail Financial Innovation and Stock Market Dynamics: the Case of Target Date Funds](#), NBER Working Paper 28028

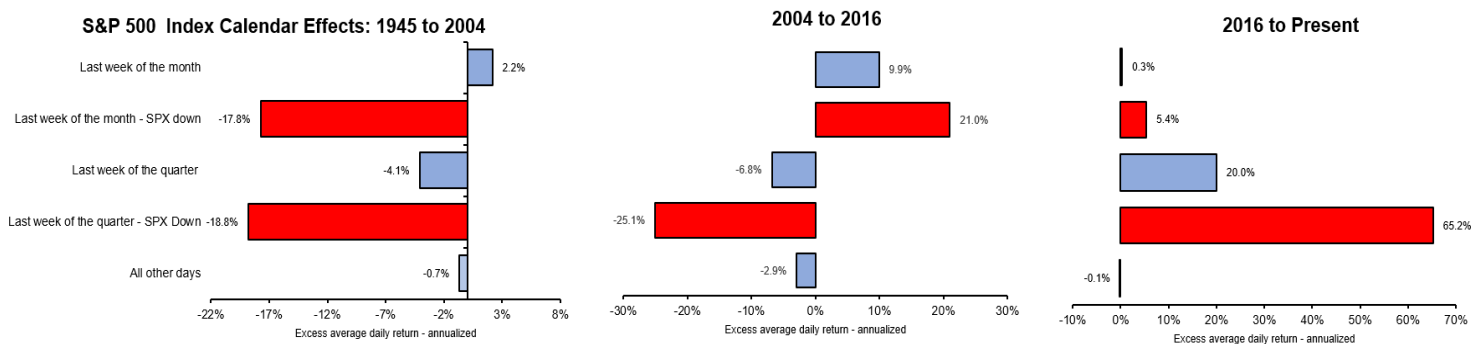


Effects

My hypothesis is that **the growth of target-date funds into a \$3 trillion juggernaut must have had at least some effect on the structure of capital markets**. Seasonal patterns are a first area to investigate. If my theory is correct, the rise of target-date funds should cause assets' performance to be more mean-reverting as "the whale" must rebalance into the lagging asset classes. The TDF whale hypothesis predicts that **equities should display abnormal returns at the end of the month or the end of the quarter following stock market sell-offs**. This effect must have grown over time as TDF assets rose from almost nothing in 2004 to about \$3 trillion today.

The chart below shows precisely that. The bar represents the excess return of the S&P 500 index around rebalancing periods. For example, the top bar measures the difference between the S&P 500 index' annualized average return in the last week of the month and its annualized return over the sample period.

I am most interested in the red bars, which measure excess return at the end of the month / quarter when stocks' returns were negative over the past three months – the trigger for TDFs' rebalancing flows. This return was negative in the pre-TDF era between 1945 and 2004. Academics have documented that stocks' tend to display positive momentum because most retail investors and commodity-traded advisors chase trends. The pattern changed in the infancy of the TDF industry: returns were abnormally positive in the last week but remained abnormally negative in the last week of the quarter between 2004 and 2016. The pattern has fully inverted since 2016: **the S&P 500 index has returned an insane 80.6%² annualized in the last week of quarters when stocks' trailing 90-day return was negative**.



I can already hear the outraged screams of my quant-minded Twitter followers: "Vincent just annualized returns observed over just five weeks! Chart crime! Burn him and cancel his reports for all eternity!"

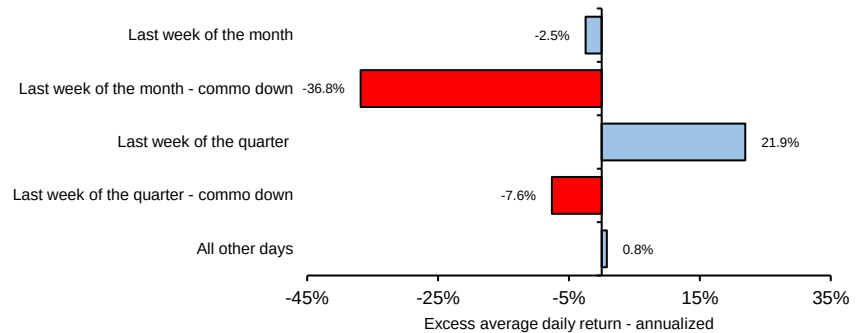
I plead for clemency. First, it is not my fault that there have only been 24 quarters since January 2016. Second it is not my fault that stocks were up in the last week of most of these quarters (I would argue that TDFs and index funds may have something to do with that, but that would distract the jury), leaving only 24 usable observations out of a theoretical total of 120.

² This translates into an annualized excess return of 65.2% as stocks rallied by 15.4% annually over the period. Jesus.

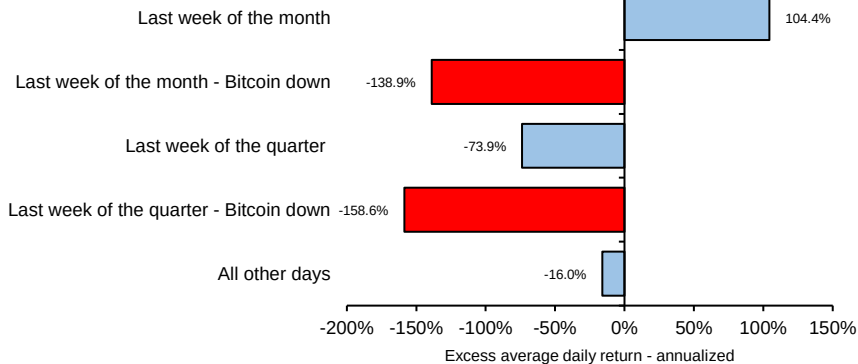


Third, I would like to introduce three new witnesses to back my claim. First, the Bloomberg commodities index, which is not impacted by TDF flows. Commodities have lost an annualized 38% in the last week of months when their trailing 90-day return was negative since 2016. Returns on the last week of the quarter in down quarters were also negative, albeit less so. In other words, **return momentum persisted in commodity indices** – which are traded by trend-following CTAs, rather than rebalancing TDFs.

Commodities Calendar Effects: 2016 to Present



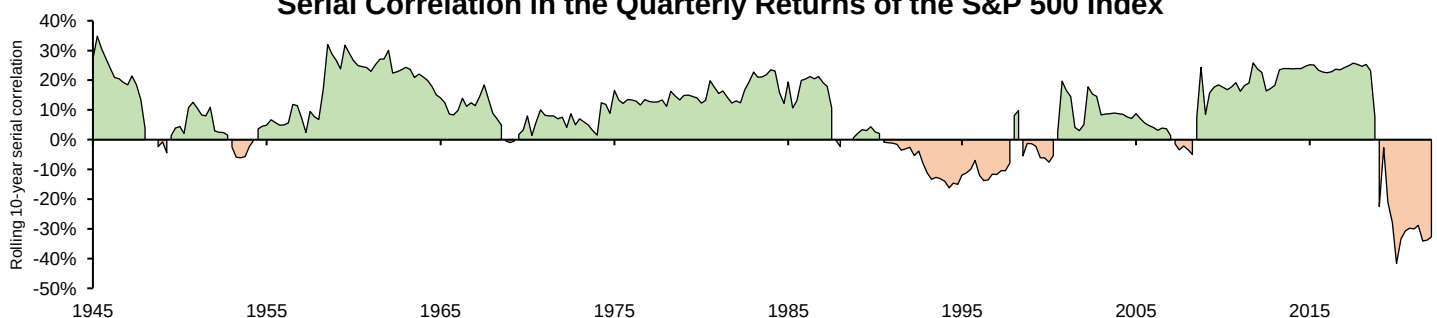
Bitcoin Calendar Effects: 2016 to Present



Bitcoin will be my second witness. Unfortunately (or fortunately? Time will tell) for investors, the DOL does not consider cryptocurrency as an acceptable asset for QDIAs, which invest exclusively in stocks and bonds. Like commodities, **the last week of months and quarters when bitcoin lost value over the past 90 days was deeply negative**. I am not going to quote the actual performance because annualizing bitcoin's daily returns is indeed chart crime.

Time for witness #3, the serial correlation of non-overlapping quarterly S&P 500 index returns since 1945. As shown in many academic studies, stocks' quarterly returns have tended to be positively correlated. Gains beget more gains, and losses beget more losses. **But stocks' quarterly returns became mean-reverting in the 2010s, just as the target-date fund whale appeared**. The serial correlation in the quarterly returns of the S&P 500 index fell to *minus* 33% in the past ten years. The lowest value observed in the pre-TDF era was just minus 16%. I rest my case.

Serial Correlation in the Quarterly Returns of the S&P 500 Index





Implications

So what? Does it matter that stocks tend to rally in the last week of the quarter? Is there any reason to think that momentum is better than mean-reversion in stocks' returns?

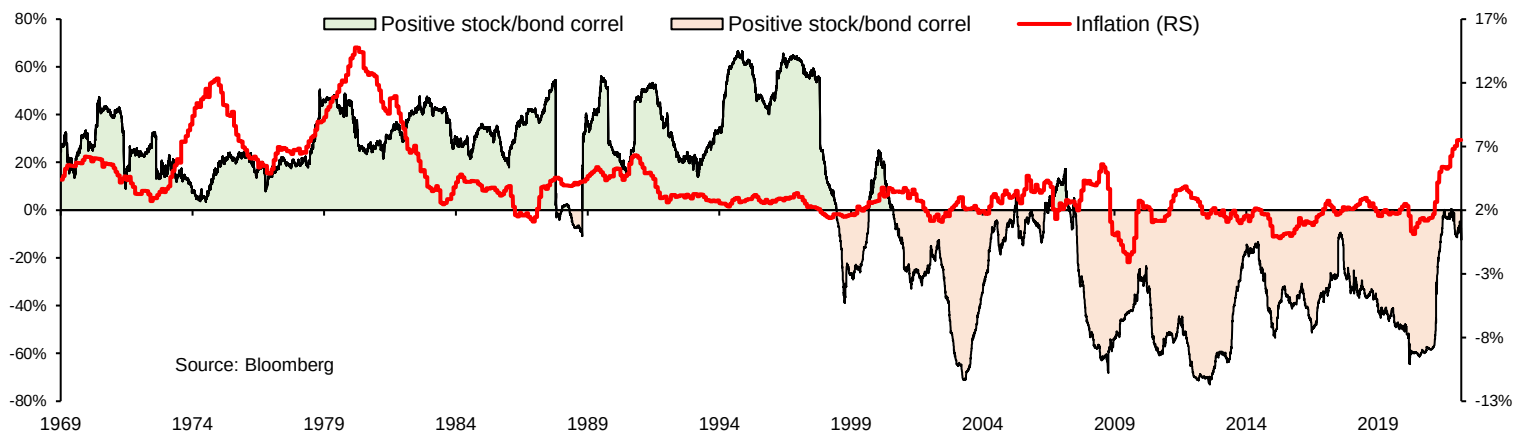
I tend to agree that seasonal patterns are irrelevant for most investors. **Quantitative hedge funds will surely exploit this anomaly and smooth these effects over time.** A more virtuous person than me may worry that this process would transfer wealth from workers' 401Ks to the pocket of short-term arbitrageurs, but I work for a broker-dealer and clients' trades ultimately pay for my boys' school tuitions.

On the other hand, I think that **TDFs' impact on stocks' returns and their correlation with bonds matters a lot for society and monetary policy.**

First, the fact that stocks always bounce back when **the TDF whale bails out the stock market at the end of the quarter has induced risk-taking.** "Stonks only go up" became the mantra of the generation which started investing in the 2010s. Buying the dip has indeed worked in the past decade: corrections were brief and shallow, and, as a result, valuations never normalized. The counterfactual is impossible to prove, but I suspect that the TDF whale is at least partly responsible for the fact that the S&P 500 index trades for 35 times cyclically-adjusted earnings when inflation is running at 7.5%, the Fed is tightening in a hurry, and Russia is threatening Europe with supersonic nuclear missiles. My guess is that investors will eventually buy one too many dip, which will turn into a savage bear market.

Second, if my theory is correct, **the TDF whale forces stocks and bonds to move in opposite directions irrespective of the economic environment.** Cross-asset correlations had an economic meaning in the pre-TDF era. When inflation shot above the Fed's 2% target in the 70s, stocks and bonds' returns were positively correlated, as investors rightly anticipated that rate hikes would harm both stocks and bonds. This correlation turned negative after the great moderation of inflation in the mid-90s. Because deflation was the dominant fear, falling bond yields (and rising bond prices) indicated economic pessimism, which hurt stocks' returns. Stocks and treasuries moved in opposite directions in the low-growth / low inflation era of the New Normal.

Trailing 1-Year Correlation between the S&P 500 Index and 10-Year Treasuries





This era persists today: the rolling one-year correlation between the returns of the S&P 500 index and that of 10-year Treasuries is *minus* 15%. A ghost of the 70s coming back to the 2022 stock market would surely be confused by investors' current behavior. Inflation is running at 7.5% and may rise above 10% in March. The Fed has made a brutal and unexpected hawkish pivot. The Russia/Crimea war is a greater shock to global commodities market than the 1973 oil crisis.

And yet, **investors flock into bonds when stocks drop, even if corrections are caused by fears of inflation, which the Fed said will be met with higher rates!** The inmates have taken over the asylum.

Last, **the TDF whale may influence monetary policy by boosting the wealth effect transmission channel.** Let me explain. When the Fed cuts rates, it increases the price of bonds. Because bond prices are rising, TDFs must buy stocks to maintain their policy weights. As a result, stock prices rise faster than they would otherwise have, people are richer, and a stronger wealth effect magnifies the impact of monetary easing.

That sounds pretty good – unless one cares about the impact of the wealth effect on inequalities, but the Fed does not consider that, otherwise it would never have invented the term “wealth effect” in the first place (but I digress).

Unfortunately, this effect is a double-edged sword. When the Fed starts to hike, bond prices will fall. As a result, TDFs will need to sell stocks to return to their policy weights. In other words, **TDFs will magnify the wealth effect in both directions**, which should result in more economic volatility.

Moreover, TDF rebalancing flows may lead the Fed's economists to misinterpret the signals sent by the bond market. **If the Fed mistakenly assumes that TDFs rebalancing flows convey meaningful data about investors' economic expectations, it may dismiss persistent inflationary prints as “transitory”.** For example, the 2020-2021 stock market rally forced TDFs to buy loads of bonds because they underperformed stocks so much. As a result, yields did not rise as much as they should have given the rapid increase in inflation. The Federal Reserve saw these low yields as a proof that the bond market – which we all know is blessed with perfect foresight and infinite wisdom – thought that inflation was indeed transitory. As a result, it ignored the voices of [early inflationistas such as this analyst](#) and finds itself in the horrible position of having to tighten amidst a slowing economy, a global oil shock, and a possible nuclear war.

Due to the complexity of capital markets, efficient markets hypothesis zealots can always dismiss my hypothesis that the tail of passive TDF flows is wagging the dog. But this report presented a preponderance of evidence, from weird end-of-month patterns to the disappearance of the momentum anomaly and changes in cross-asset correlations.

Moreover, **the macro-economic consequences of my hypothesis have all been verified:** rarer and shallower corrections which never last more than a quarter, general “buy the dip” mentality, permanently rising valuations despite deteriorating fundamentals, a greater reliance on the wealth effect, the dismissal of widespread inflationary pressures, and an increased probability of a monetary policy blunder.

It is time for policymakers and investors to study and understand the impact of the TDF whale.



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